Zayaan Bhanwadia

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Education

University of Toronto — Sept. 2025 – Dec. 2028
Hon. B.Sc. in Computer Science, Software Engineering Specialist (Co-op) | GPA: 3.7/4.0 — *Toronto, ON*
Activities: 3x Hackathon Winner, VP of Technology @ Computer Science Union, Web Developer @ UTRA

Experience

Full-Stack Software Engineer Intern *(Incoming)* — Sept. 2026 – Dec. 2026
FreshBooks — *Toronto, ON*

- Incoming co-op developing Python backend microservices and React UI for a platform serving **10M+ users**.
- Will ship production code on a product team, deploying with Docker, Azure DevOps, and Google Cloud Platform.

Autonomy Software Developer *(Mapping & Planning)* — July 2025 – Present
Formula Student Driverless – University of Toronto Formula Racing — *Toronto, ON*

- Developed mapping and path-planning software in C++/ROS2 for **Canada's first driverless FSAE car**.
- Built a FastSLAM and GraphSLAM pipeline fusing **6 sensors** to localize the car within **15 cm** of optimal racing line.
- Implemented Delaunay-based centerline and raceline planning to output low-latency, real-time trajectories to controls.
- Validated and optimized algorithms across **20+** simulation and rosbag-replay runs to catch failures before track tests.

Software Engineer Intern *(Backend AI)* — July 2026 – Present
FlyRank AI — *Toronto, ON*

- Built a centralized embeddable widget platform, routing form submissions to a dashboard via a one-line script tag.
- Secured a public endpoint with validation, rate limiting, spam filtering, and a geolocation fallback, passing **80+ tests**.
- Prevented duplicate charges under **50 concurrent requests** by deduplicating billable events at the database level.
- Synced Stripe payment updates through secure, idempotent webhooks, eliminating duplicate records and manual fixes.

Full-Stack Software Developer Intern — May 2026 – July 2026
KorraNet Creative (Riipen Program) — *Remote*

- Optimized Meta OAuth token handling and storage, reducing token refresh failures by **40%+** for AI integrations.
- Reduced system crashes and failed API calls by **85%+** through adding API error handling and timeout prevention.
- Migrated hardcoded secrets to Firebase Secret Manager and implemented real-time secret ingestion for API calls.

Projects

Embedded Key-Value Store | *C++, POSIX Threads, Linux*

- Built an embedded C++ database with data sharding to prevent global lock bottlenecks and improve throughput.
- Implemented shared locks for non-blocking reads across **4 threads**, preventing race conditions during writes.
- Designed data persistence by writing updates to a log file before memory to ensure crash recovery and durability.
- Created a file compaction routine to safely clean up keys, **reducing disk usage by 45%+** without pausing the app.

Context Sync Extension | *TypeScript, VS Code Extension API*

- Developed a VS Code extension with **500+ downloads** that automates AI chat context sharing across environments.
- Designed a graph-based Markdown schema to maximize context density per token across AI assistant sessions.
- Integrated GitHub Copilot, OneDrive, and Google Drive to automate syncing and enable one-click AI workspace setup.

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C++, C, SQL, Bash, HTML/CSS
Libraries/Frameworks: FastAPI, Express.js, Node.js, SQLAlchemy, React.js, ROS2, OpenCV, YOLOv11
Developer Tools: Linux, Docker, Git, GitHub, Claude Code, PGAdmin, VS Code, Jira/Confluence
Cloud/Databases: PostgreSQL, MySQL, Google Cloud Platform, IBM Cloud, Vercel, Vultr, Railway, Supabase